

A new comprehensive filtering model for pump shut-in water hammer pressure wave signals during hydraulic fracturing

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ABSTRACT

In the process of fracturing, the position of the fractures can be identified by analyzing the spectrum of the water hammer pressure wave signals after pump shut-in. However, the collected water hammer pressure data contain a lot of noises, which brings difficulty to the spectrum analysis and affect the accuracy of the final results. Removing multiple kinds of noises while protecting useful fracture response is the challenge. However, the filtering model for the water hammer pressure wave signals is absent. This paper proposes a new comprehensive filtering model for pump shut-in water hammer pressure wave signals. The filtering model contains three parts. First, we designed a two-step filtering method, including Hampel filter and FIR low-pass filter, to filter signals in the main channel. Second, multi-frequency-adaptive notch and median filter were combined to construct the signals in the reference channel, which can be used to protect the useful frequency of the water hammer wave signals and pressure response of the fractures from being eliminated. Third, the filtered signal is obtained by adaptive cancellation between the signals in the main channel and the reference channel. After that, the filtering effects of signals with and without fractures were studied by constructing simulated signals and collecting the signals in the laboratory experiment. The signal-to-noise ratio (SNR) gain and the mean square error (MSE) reduction were selected as indicators for the model validation, and the cepstrum was also used as the indicator to verify the preservation of fracture response characteristics. Besides, the noise adaptive range of the model was evaluated by increasing the noise intensity to the constructing signal. This research can provide a new and useful filtering method for water hammer pressure wave signals in the field.

1. Introduction

Hydraulic fracturing, as an efficient stimulation technology, has been widely used in various oil and gas reservoir reconstruction and achieved remarkable results (Li et al., 2020). A large number of production practices have proven that hydraulic fracturing is the key technology for unconventional reservoir reconstruction and efficient development (Cong et al., 2021). It is an effective way for the development of the unconventional oil and gas reservoir (Wang et al., 2018a,b; Cong et al., 2022). High viscosity fracturing fluid were pumped through the wellbore into the reservoir by the high-pressure pumps on the surface. Then fractures were generated, creating new fluid channels between the reservoir and the wellbore (Adachi et al., 2007). Whether fractures are open in all the designed perforation position is an essential issue in hydraulic fracturing. The fracture locations (or the flow entrance) are a critical parameter in hydraulic fracturing evaluation. Therefore, analysis and evaluation of the fracture location is a necessary requirement of

hydraulic fracturing diagnostics. Water hammer fracture diagnostics has been developed and validated in the field application since 2016 (Bogdan et al., 2016; Korkin et al., 2020). Fracture diagnostics based on water hammer pressure wave signals can identify the fracture locations by analyzing the spectrum characteristics of water hammer signals, which has advantages of low cost, easy operation, and great real-time performance (Bogdan et al., 2016; Klem et al., 2017; Kabannik et al., 2018).

The principle of water hammer fracturing diagnostics is as follows. When the pumps shut down during the hydraulic fracturing, the sudden change of the fluid flow rate at the wellhead causes rapid pressure change inside the wellbore, known as “water hammers” (Connor et al., 2018). Water hammer signatures are caused by tube waves traveling down to the wellbore from the surface, meeting completion components (such as tube end, fractures, tube expansion, or a combination of all these factors) and then traveling back (Parkhonyuk et al., 2018). Simply put, a water hammer is a pressure wave that transmits throughout the

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well hydraulic system. In the process of transmission, the pressure wave is attenuated due to the energy loss. Water hammer characteristics are determined by the shut-in process, wellbore geometry, fluid properties in the well, hydraulic fracture geometry, and rock physical properties around the fractures (Ma et al., 2019). In previous studies, the interpretation of water hammer signatures can be used as a simple diagnostic tool to characterize hydraulic fractures. The collected water hammer pressure wave signals contain the pressure response characteristics of fractures (the water hammer wave propagates along the pipeline and is reflected by the hydraulically connected fractures, resulting in sudden changes of ground pressure and carrying fracture information).

However, various fluctuation events, such as pipeline vibration and fracture opening, are unavoidable during fracturing and pump shut-down. The water hammer pressure wave signals collected in field contains a lot of spike noises, random noises, and fixed-frequency noises. The existence of these noises will substantially interfere the time-frequency analysis and the pressure response analysis of the water hammer pressure wave signals, thus affecting the accuracy of fracture localisation. Therefore, how to use reasonable filtering methods to filter out various of noises is a crucial problem in water hammer fracturing diagnostics. At the same time, it is necessary to protect the fracture information in the signals. This process requires a more accurate filtering model.

At present, the research on filtering methods in the oil industry mainly focuses on seismic exploration, wireless measurement while drilling, and oil and gas production prediction. In seismic exploration, an adaptive method is a very effective filtering method. Liang and Mike, (2009) proposed an adaptive FK filtering algorithm, which can remove noises surface micro-seismic data. The algorithm mainly combined the geometric features of surface seismic sequence with the distribution features of micro-seismic events. Then it improved the SNR of micro-seismic signals by superposition processing method. Liu et al. (2012) suppressed random noises in seismic signals by combining EMD with adaptive thresholds for translation and scale invariance. Song et al. (2013) proposed an improved method of adaptive estimation filtering, considering the addition of constraints to make the iterative method move towards the direction of expected convergence. Peng et al. (2019) adopted an adaptive bilateral filtering method based on multiple windows to suppress the noises in the complex fault block area, which could better maintain the edge information. In the wireless MWD system, the adaptive filtering method and its improved algorithm combined with other methods are commonly used. Li and Reckmann (2009) proposed a filtering method based on the adaptive algorithm. The fundamental frequency of pump noise was taken as the frequency of the reference signal to realize effective filtering. Qiao et al. (2016) proposed a new filtering algorithm by analyzing the noises of mud pulse signals. An adaptive notch was used to reduce the correlation between the signals in the main channel and the reference channel, thus protecting the helpful signal from being canceled. Peng (2018) proposed a signal processing method combining adaptive notch filter and adaptive filter. The adaptive notch filter filtered the valuable components, and then the output of the notch filter was used as the reference signal of the adaptive filter. Li (2019) studied noise reduction methods from the perspectives of adaptive denoising method based on pump-impact interference model and zero-space tracking. In oil and gas production prediction, Kalman filter is often used in many prediction methods. Kalman filter is also a special adaptive method. Wang and Li (2010) proposed an improved ensemble Kalman filtering method, which can avoid the inconsistency caused by the dynamic field update. Li (2012) used a semi-iterative ensemble Kalman filter to solve the problem that dynamic reservoir field is not consistent with reservoir model in nonlinear problems, which can accurately invert nonlinear parameters. Liu et al. (2020) preprocessed injection-production data using Kalman filtering algorithm and nonlinear diffusion filter respectively. This process reduces interference with the machine learning model.

In these research fields, some filtering ideas are effective

combinations of several filtering methods. The adaptive approach is one that is often considered. The limitations of previous combined filtering methods for water hammer pressure wave signals are discussed as follows. Firstly, spike noises significantly influence the pressure response information in cepstrum analysis. Many of these combined filtering methods do not consider eliminating spike noises interference. Secondly, how to protect the pressure response of the fracture (which is shown as small fluctuations at certain moments on the pressure wave curve) should be taken into account in the filtering of the water hammer pressure wave signals. Otherwise, it will easily lead to misjudgment and affect fracture location identification. It is a large challenge to remove a variety of noises while protecting useful pressure response of fractures. Hence, it is essential to develop a comprehensive filtering method for water hammer pressure wave signals.

This paper proposes a new comprehensive filtering model for water hammer pressure wave signals, Fixed Parameter Filter-Adaptive Filter (FPF-AF). The comprehensive filtering model is based on the adaptive noise cancellation. Firstly, we proposed a two-step filtering method for the spike noises and high-frequency noises in the main channel of the adaptive noise canceller, that is, the filtering method combining Hampel filtering and FIR low-pass filtering. The model ensures that the noises obviously different from the useful signal can be filtered out efficiently. At the same time, the input quality of the signal in the main channel is guaranteed. Secondly, aiming at the difficulty of obtaining the noises in the reference input, we innovatively proposed a construction method in the reference channel based on the Multiple Frequency Adaptive Notch-Median Filter (MFAN-MF). The multi-frequency adaptive notch is adopted to remove the oscillation frequency of irregular sine wave or irregular square wave. The median filtering to remove baseline drift is introduced to filter excessive trend fluctuation caused by pressure response of the fractures. At the same time, the noise trend can be removed to make the amplitude fall to zero fluctuation. Finally, the noise constructed by the MFAN-MF method is input into the adaptive noise canceller, which is cancelled with the signal obtained by the two-step filtering method. Simulation signals and laboratory signals verify the effectiveness and feasibility of the comprehensive filtering method. This research can provide a new and useful filtering model for water hammer pressure wave signals in the field.

2. Signal characteristics and method analysis

There are many kind of noises in the water hammer pressure wave signals, including spike noises, random noises, fixed frequency noises close to the frequency of the useful signal, and fixed frequency noises which are much different from the useful signal frequency. Spike noises refer to the abnormal burr, jump, and other interference in the signal due to the working environment influence and the device manufacturing process. Random noises mainly refer to the noises whose frequency and propagation direction have no fixed rules. The fixed frequency noises refer to the noises whose frequency is fixed and will not change. The fixed frequency noises generated in the fracturing process may be the noises that are significantly different from or close to the frequency of the water hammer pressure wave signals. This paper proposes a new comprehensive filter model of Fixed Parameter Filter - Adaptive Filter.

2.1. Filtering model development

First of all, considering that the noise amplitude is large and the types of noise are many, and the pressure response of the fractures need to be protected, it is best to use a simple and effective adaptive filter for denoising. Our model is based on adaptive noise canceller. Generally, according to the basic structure of adaptive noise canceller and considering the characteristics of water hammer pressure wave signals, our model can be divided into three parts: (1) Main input channel: A two-step filtering method for the main input signal; (2) Reference input channel: A Multiple Frequency Adaptive Notch - Median Filter (MFAN-

MF) method for the reference input signal. (3) Adaptive noise cancellation between the main input signal and the reference input signal. We will present the derivations as follows.

Main input channel: A two-step filtering method for water hammer signal with noise. There are many types of noise in the collected signal. We can initially filter the noises which can be easily separated from the useful signals. Generally, smooth filtering is carried out first. We need to design a suitable filtering scheme to filter out the spike noises. At the same time, high-frequency noises are removed within the desired accuracy and acceptable time limits. In addition, the signal processing in the previous step should not be distorted to affect the signal processing in the later step. Considering the characteristics of water hammer pressure wave signals, we design a two-step filtering method for initial filtering. Firstly, the Hampel filter is selected to filter the spike noises, and then the FIR (Finite Impulse Response) filter is used to filter the high-frequency noises.

The acquired signal $r(t)$ generally contains noises caused by system or environment instability, which will produce a lot of spike noises. The commonly used method to remove the spike noises is the median filter. Although this method can filter out the abnormal spike noises, it may also delete many original signal data points. The filter closely related to the median filter is the Hampel filter. The Hampel filter works like a median filter, but it only replaces values a few standard deviations away from the local median value (Jiang et al., 2011). This filter can help remove outliers from the signal without over-smoothing the data. For a given sample of data x_s , the algorithm: 1. Centers the window of odd length at the current sample. 2. Computes the local median m_i , and standard deviation σ_i , over the current data window. 3. Compares the current sample with $n_\sigma \times \sigma_i$, where n_σ is the threshold value. If $|x_s - m_i| > n_\sigma \times \sigma_i$, the filter identifies the current sample x_s as an outlier and replaces it with the median value, m_i . Consider a frame of data that is passed into the Hampel filter. After smoothing by the Hampel function, the signal $p(t)$ is obtained. And the spike noises are filtered out.

After removing the spike noises, the signal $p(t)$ contains many fixed frequency noises, which is wildly different from the useful signal frequency. In order not to cause too much influence on the subsequent analysis, we use FIR low-pass filter to filter the high-frequency noises. The FIR low-pass filtering method can effectively filter out part of the high-frequency noises. The full name of the FIR filter is a finite impulse response filter. Because the filter has good linear phase characteristics, it has been widely used in digital signal processing (Huang, 2019). Frequency response of the FIR low-pass filter has three ranges: passband, stopband, and transition band. The frequency response is close to 0 in the stopband and close to 1 in the passband (Shao et al., 2017). If the response of its unit sample is a finite sequence of points $M+1$, and the input signal is the convolution of $p(n)$ and $h(n)$, that is:

$$d(n) = \sum_{k=0}^M p(k)h(n-k) = p(n) * h(n) \quad (1)$$

Where “*” represents the convolution operation, $h(n)$ is the response of unit sampling, and $p(n)$ is the input signal of the FIR low-pass filter. The transfer function of this filter can be expressed as:

$$H(z) = \sum_{n=0}^M h(n)z^{-n} = h(0) + h(n)z^{-1} + \dots + h(M)z^{-M} \quad (2)$$

where $H(z)$ is the transfer function. After processing such a filter, the signal $d(t)$ is obtained, which filters out part of the high-frequency noises.

Reference input channel: A Multiple Frequency Adaptive Notch - Median Filter (MFAN-MF) method for the noise input signal. When we use adaptive noise cancellation method to filter, the signal in the reference channel is the noise signal. It is difficult to select the input signal of the noise channel. The noise can be extracted from the signal in the main channel as the reference signal, but there are the following

problems: because the signal in the main channel contains useful signals, when some useful signal components are input to the reference channel, the useful signals in the final output will be partially eliminate. Therefore, we need to separate the useful information in the noise channel in order to protect the useful signals from being eliminate. Finally, we propose a MFAN-MF method for reference channel.

For the water hammer pressure wave signals, the oscillation frequency is one of the useful information needs to be protected. Therefore, filtering the useful oscillation frequency in the reference channel is necessary during the noise signal construction. A multi-frequency adaptive notch is adopted. The signal $d(t)$ is input into the multi-frequency adaptive notch filter and cancels with the input reference signal $x(t)$ several times. Until you get a noise $v(t)$ that does not contain the useful signal frequency. The frequency value of this reference signal $x(t)$ is set to the useful frequency value. The adaptive notch filter has two outputs: filter output and notch output. Compared with the ordinary digital filter, the adaptive notch filter has a narrower stopband and passband (Wang et al., 2018a,b). The out-of-band attenuation is faster. Therefore, it can realize narrow-band filtering and narrow-band notch. Recursive process of adaptive notch filter:

$$\begin{cases} y(k) = \omega_c(k)x_c(k) + \omega_s(k)x_s(k) \\ v_1(k) = d(k) - y(k) \\ \omega_c(k+1) = \omega_c(k) + 2\mu v_1(k)x_c(k) \\ \omega_s(k+1) = \omega_s(k) + 2\mu v_1(k)x_s(k) \end{cases} \quad (3)$$

where $x_c(k)$ and $x_s(k)$ are the two reference signals, $\omega_c(k)$ and $\omega_s(k)$ are the weight coefficients of the two reference signals, μ is the step factor, $d(k)$ is the input signal, $y(k)$ is the filtering output, and $v_1(k)$ is the output after the first notch. The multi-frequency adaptive notch filter adopts a cascade structure. The output of the first stage notch filter is the input of the second stage notch filter. The zeros and poles of the cascaded system correspond to the zeros and poles of each notch filter respectively. we can directly change adjustment parameters of one stage without affecting the other stage of the notch filter (Liu, 2008). It has the advantages of the single-frequency notch filter and makes the performance adjustment simple and effective (Chen, 2016). The useful frequency in the water hammer pressure wave signal is not necessarily a single certain frequency, at this time, it is necessary to go through several notch filters for processing until these frequency components are removed. The number of the multi-frequency adaptive notch filter is determined according to the frequency component of the useful signal. The output of the upper notch filter is the input of the next notch filter. After a multi-frequency adaptive notch operation, the output can be obtained as $v(t)$.

The pressure response of the fracture showing small fluctuations in the pressure curve is another useful information that needs to be protected. Median filter is adopted. In addition, considering that the amplitude of the constructed noise should be near zero, we introduce the idea of removing baseline drift to de-trend the signal. Combined with the previous step, the median filtering to remove baseline drift can be directly adopted, which makes the filtering operation simpler and improves the efficiency of filtering. The median filter is a nonlinear digital filtering technology that can suppress noises and protect the edge. The basic idea of using the median filtering method to remove the baseline drift is as follows: first, removing the more enormous value in the signal to obtain the trend signal containing only the baseline. Then superimpose it with the original signal to eliminate the baseline drift interference in the original signal. The median filtering principle: the value of a point in the signal is replaced by the median value of each point in a neighborhood of the point (Xu et al., 2012). The application of median filtering to baseline drift has two functions: one is to remove excessive trend fluctuation at the pressure response of the fracture. And the other is to make the amplitude of the resulting noise fluctuate around zero. Here, the input signal is the output signal of the adaptive notch filter, and the size of the sliding window of the median filter is $L = 2R + 1$, $L < N$ (Zhao et al., 2012). Then the signal processed by the median filter is:

$$n(t) = \begin{cases} v(t) - \text{Med}[v(1), v(2) \dots v(t+R)] & (1 \leq t \leq R) \\ v(t) - \text{Med}[v(t-R) \dots v(t) \dots v(t+R)] & (R \leq t \leq N-R) \\ v(t) - \text{Med}[v(t-R), v(t-R+1) \dots v(N)] & (N-R \leq t \leq N) \end{cases} \quad (4)$$

where $n(t)$ is the signal processed by the median filter, $v(t)$ is the adaptive notch filter's output signal, R is the neighborhood size around a certain point. N is the signal length.

Adaptive noise cancellation between the main input signal and the reference input signal. After the above two parts of processing, we get the signal in the main channel and the reference channel. Finally, we can filter the residual noises by adaptive noise canceller. The noise $n(t)$ without the signal frequency is input into the adaptive noise canceller, which cancels with the signal $d(t)$ after filtering out the high-frequency noises. And finally, the filtered signal $e(t)$ is obtained. The core of the adaptive noise canceller is the adaptive filter, which is controlled by the adaptive algorithm to achieve the best filtering. Here, one input of the adaptive noise cancellation system is the signal $d(t)$ after filtering part of the high-frequency noises through the FIR low-pass filter. The other input is the noise $n(t)$ obtained after the median filter to remove the baseline drift, which is related to the noises contained in the original signal. The principle of adaptive noise cancellation is as follows (Ma and Cheng, 2016). It is assumed that the original input end of the adaptive noise cancellation system be $d_j, d_j = s_j + n_0$, where s is the useful signal, n_0 is the noise to be canceled and is not related to s . The reference input end is n_1 , where n_1 is the noise related to n_0 but unrelated to s . And the output of the system is $e_j, e_j = d_j - y_j$. Here, the filter's transmission function can be adjusted automatically according to the system's output signal. Assuming that s, n_1 , and n_0 are all stationary random processes with zero mean:

$$e_j = d_j - y_j = s_j + n_0 - y_j \quad (5)$$

where e_j is the output of the adaptive noise canceller, d_j is the original input signal, and y_j is the filter output. The mean square value of the output signal is expressed as:

$$\begin{aligned} E[e_j^2] &= E[(d_j - y_j)^2] = E[(s_j + n_0 - y_j)^2] = E[s_j^2] \\ &= E[(n_0 - y_j)^2] + 2E[s_j(n_0 - y_j)] \end{aligned} \quad (6)$$

Since s is not correlated with n_0 and n_1 , so s is not correlated with y_j , then:

$$E[e_j^2] = E[s_j^2] + E[(n_0 - y_j)^2] \quad (7)$$

where $E[s_j^2]$ represents the power of the signal. As can be seen from the above expression, if you want the output signal to contain only useful signals or the mean square value of the output signal to be minimum, $E[(n_0 - y_j)^2]$ it must obtain a minimum value. According to Eq. (6), the equivalence condition is that $E[(e_j - s_j)^2]$ is required to obtain the minimum value. The mean square value of the error between the output signal and the useful signal is required to be minimum. The error output e of the system is the best approximation of the useful signal s .

2.2. Filtering process

The process of the comprehensive filtering method shows in Fig. 1. As discussed above, we can divide all the process into three parts: (1) Main input channel: A two-step filtering method for the main input signal; (2) Reference input channel: A MFAN-MF method for the noise input signal. (3) Adaptive noise cancellation between the main input signal and the reference input signal. The collected signal $r(t)$ contains spike noises, random noises, low-frequency noises and high-frequency noises, and we hope to obtain accurate physical parameters from the signal.

In part 1, we can perform a two-step filter for the easily separated spike noises and the high-frequency noises. Firstly, Hampel filter is used to filter out the spike noises, and the filtered signal is denoted as $p(t)$. The filter not only helps to remove the spike noises, but also does not over-smooth the data. Then the FIR filtering algorithm is used to filter out the high-frequency noises and get the signal in the main channel $d(t)$. In part 2, we need to separate the noise from the signal $d(t)$ by using the noise channel construction method. The main channel input of the adaptive notch is $d(n)$, the input of the reference channel is the sinusoidal signal $x(t)$ whose frequency is equal to the oscillation frequency, and the output is the signal $v(t)$ without the oscillation frequency. The signal $v_1(t)$ is obtained after median filtering of the signal $v(t)$. The slight fluctuation of the pressure response in $v_1(t)$ is smoothed out. Then $v(t)$ minus the smoothed signal $v_1(t)$ to de-trend, which is used as the reference signal of the adaptive noise canceller, denoted as $n(n)$. $n(n)$ almost does not contain useful signal. In this way, the useful signal can be well protected in the output signal $e(n)$, and the accuracy of the

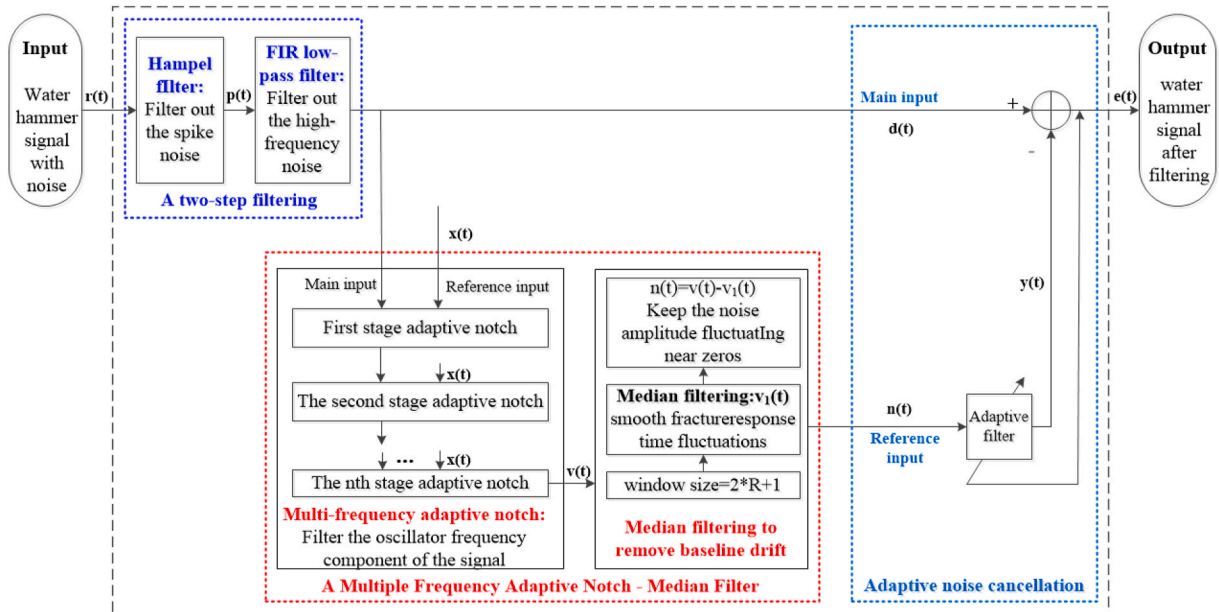


Fig. 1. New comprehensive filtering method flow.

filtering result can be improved. In part 3, the filtered signal $e(t)$ can be obtained through adaptive noise canceller. Then we can analyze the signal-to-noise ratio (SNR) and mean square error (MSE) of the water hammer pressure wave signal $e(t)$.

2.3. Model evaluation indicators

Signal to noise ratio (SNR) and mean square error (MSE). After the water hammer wave signal is filtered by the comprehensive method, the evaluation indicators are the signal to noise ratio (SNR) and mean square error (MSE). The smaller the MSE, the larger the SNR and the better the filtering effect. The SNR can be calculated by Eq. (8).

$$SNR = \frac{P_{signal}}{P_{noise}} = \left(\frac{A_{signal}}{A_{noise}} \right)^2 \quad (8)$$

where P_{signal} is the signal power, P_{noise} is the noise power, A_{signal} is the signal amplitude, A_{noise} is the noise amplitude. The SNR is generally measured in decibel (dB), which can be expressed as Eq. (9).

$$SNR(dB) = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) = 20 \log_{10} \left(\frac{A_{signal}}{A_{noise}} \right) \quad (9)$$

The mean square error (MSE) refers to the mean square of the variance between the original signal and the filtered signal. The MSE can be calculated according to Eq. (10).

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_{obs,i} - x_{mldel,i})^2 \quad (10)$$

where $x_{obs,i}$ is the original signal, $x_{mldel,i}$ is estimated signal after filtering, n is sequence number, i is serial number.

Cepstrum. Cepstrum analysis of voice and image processing are widely used in nonlinear signal processing technology. Here cepstrum is used as the indicator to verify the preservation of pressure response characteristics. Cepstrum is the inverse Fourier transform of the Fourier transform spectrum of a signal after logarithmic operation. The calculation principle of cepstrum is shown in Fig. 2. The reflection information generated by water hammer wave signals and multi-fracture reaction in fracturing process can be analyzed directly by cepstrum, and the peak value can be found by nonlinear processing method.

3. Results and analysis

3.1. Simulated pressure wave signal without fractures

The water hammer pressure wave signals containing various noises are simulated and verified to prove validity and feasibility of the method. The simulated water hammer pressure wave signals without fractures is constructed, and different noises are added to the simulated signal. The effect of comprehensive filtering method is studied, and the change of SNR and MSE is analyzed.

The details of constructing a pure signal are as follows: The sinusoidal signal with amplitude attenuation represents the normal amplitude attenuation of the water hammer pressure wave signals after the pump is stopped. Besides, the signal front part represents the pressure before stopping the pump by a supplemental 10 s signal. After a certain amplitude processing, the simulated signal can show that the water hammer pressure wave signals without fractures. On the whole, the simulated water hammer pressure wave signal is a sinusoidal signal with

amplitude attenuation (Fig. 3a). The thick arrow position indicates that the pressure begins to drop suddenly after the pump is stopped at 10 s. The simulated signal frequency is within 1 Hz, the main frequency is 0.1286 Hz, and the spectrum has a long tail (Fig. 3b).

The details of constructing the simulated signal with noise are as follows: Spike noises, random noises, low-frequency noises of fixed frequency which is close to useful signal frequency, and high-frequency noises of fixed frequency which is different from useful signal frequency are added into the pure signal. The specific process is simplified as follows:

$$r(t) = s(t) + g(t) + n(t) + q(t) + m(t) \quad (11)$$

where $r(t)$ is the signal collected by the signal acquisition system. $s(t)$ is the water hammer pressure wave signal. $g(t)$ is the spike noise. $n(t)$ is the random noise. $q(t)$ is the high-frequency noise with a significant frequency difference, and $m(t)$ is the low-frequency noise with a close frequency.

The simulated signal with noise is shown in Fig. 4a. We adopt the comprehensive filtering model to filter the simulated signal. In Fig. 4b, the red part is the first step to remove the spike noises by Hampel function. In Hampel filtering, the filtering series should be set to odd number. The filter level doesn't need to be too high, so just set it to 9. We can see that the spike noises are removed without over-smoothing the overall signal. Fig. 4c shows the signal after adopting FIR low-pass filtering in the second step, and the high-frequency noises and random noises are partially suppressed. The noises are significantly reduced. Fig. 4d is the signal processed by the adaptive noise canceller, the signal output by the comprehensive filtering model. It can be seen that the noise has been further reduced.

After the water hammer pressure wave signal is filtered by the comprehensive method, the evaluation indicators are the signal to noise ratio (SNR) and mean square error (MSE). In general, the smaller the MSE, the better the filtering effect; The larger the SNR, the better the filtering result. Comparison of SNR and MSE of signals before and after comprehensive filtering is shown in Fig. 5. Fig. 5a shows the SNR before and after the comprehensive filtering. Before the filtering, the SNR is about 27.42971 dB. After Hampel filtering, the SNR is about 29.78231 dB. After FIR low-pass filtering, the SNR is about 32.63344 dB. After the comprehensive filtering, the SNR is about 34.274 dB. And the SNR gain is about 6.84429 dB. Fig. 5b reflects the MSE before and after the comprehensive filtering. Before the filtering, the MSE is about 0.1017. After Hampel filtering, the MSE is about 0.0794. After FIR low-pass filtering, the MSE is about 0.0572. After the comprehensive filtering, the MSE is about 0.0473. The MSE decreases about 0.0544. The SNR and MSE of the signals are significantly changed after filtering. After comprehensive filtering, the SNR is higher than that of single Hampel filtering and FIR low-pass filtering, and the MSE is lower than that of single Hampel filtering and FIR low-pass filtering. The noise is reduced substantially after filtering.

3.2. Experimental pressure wave signal without fractures

To further verify the comprehensive filtering model, the signals collected in the laboratory are used for verification. We set up a test device for water hammer pressure wave signals, which disturbed the fluid in the main line through the rapid closing of the valve to generate pressure wave. In the test, the pressure in the main line (that is, the system pressure) was in the range of -0.15 MPa to 0.5 MPa, the liquid

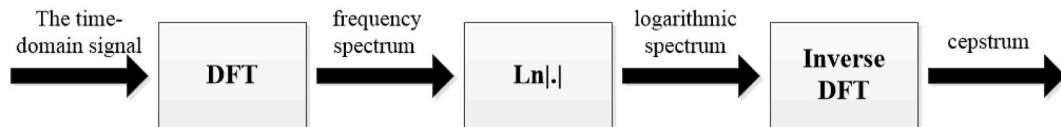


Fig. 2. The calculation principle of cepstrum.

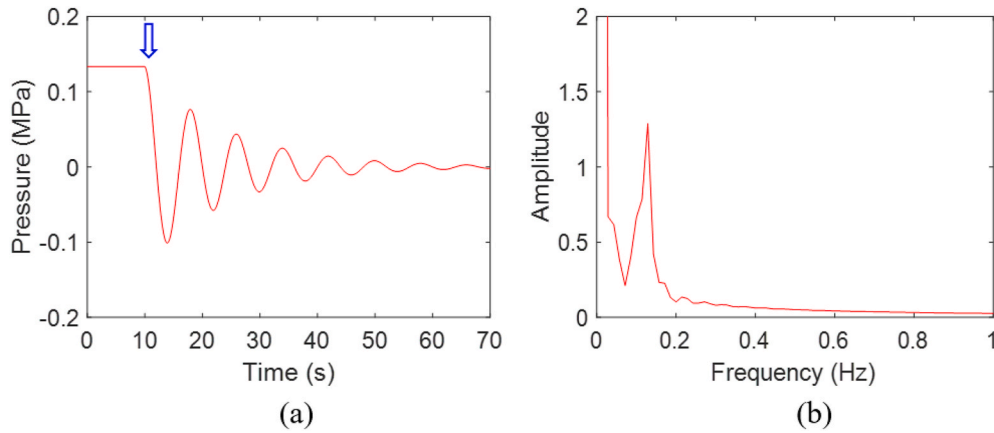


Fig. 3. Simulated signal without fractures and the signal spectrum: (a) simulated signal without fractures; (b) spectrum of the simulated signal without fractures.

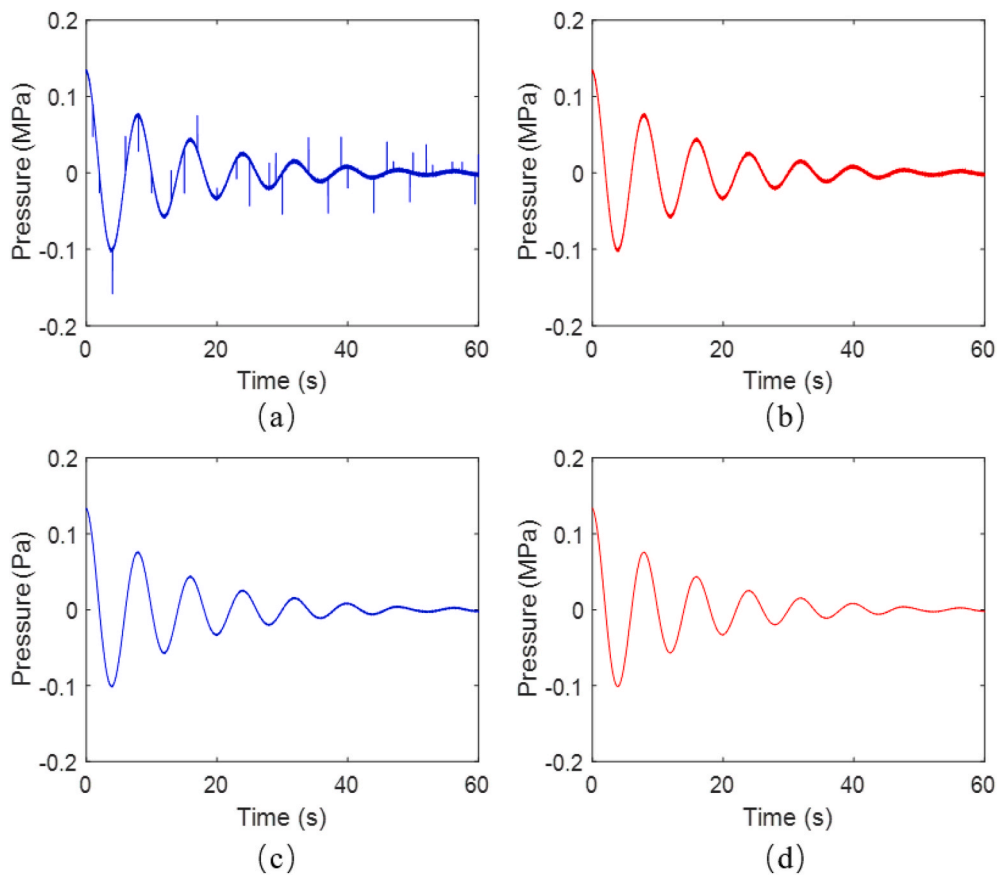


Fig. 4. Simulated signal with noise and the signals after filtering: (a) simulated signal with noise; (b) the signal after Hampel filtering; (c) the signal after FIR low-pass filtering; (d) the signal after adaptive noise cancellation.

density was 1 g/cm^3 , and the liquid temperature was 293 K. The sampling frequency is $f_s = 500 \text{ Hz}$. The signal collected in the laboratory is shown in Fig. 6a, which contains a lot of noise. The new comprehensive filtering model is adopted to de-noise. The red part shown in Fig. 6b is the signal filtered by the Hampel function, and a large number of abnormal spike noises are filtered out. The yellow part is the signal after FIR low-pass filtering, which further filters out part of high frequency noises and random noises. Fig. 6c shows the final output signal by comprehensive filtering, and it can be seen that the water hammer characteristics are well preserved.

The validity of the filtering results is further verified by spectrum analysis. The spectrum of the signal collected in the laboratory is shown

in Fig. 7a. The main frequency of the signal is about 0.21 Hz. After comprehensive filtering, the signal is shown in Fig. 7b. The main frequency of water hammer pressure wave signals is clearly displayed at the peak value, while the high frequency can be attenuated. The noise outside the useful signal band can be effectively suppressed.

3.3. Simulated pressure wave signal with fractures

The details of constructing the pure simulated signal with fractures are as follows: Firstly, a sinusoidal signal with amplitude attenuation is constructed to represent the normal amplitude attenuation of the water hammer pressure wave signals. Then, different sinusoidal signals with

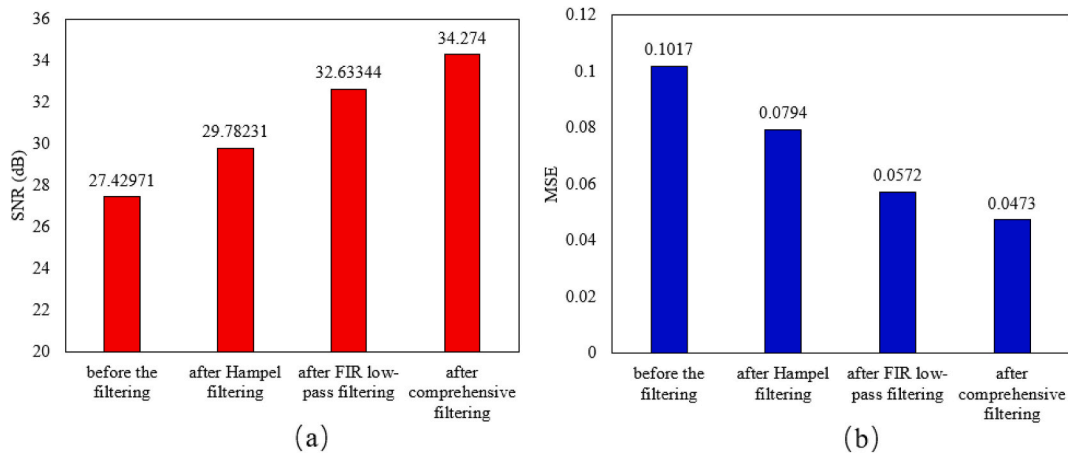


Fig. 5. Comparison of SNR and MSE before and after comprehensive filtering: (a) comparison of SNR before and after filtering; (b) comparison of MSE before and after filtering.

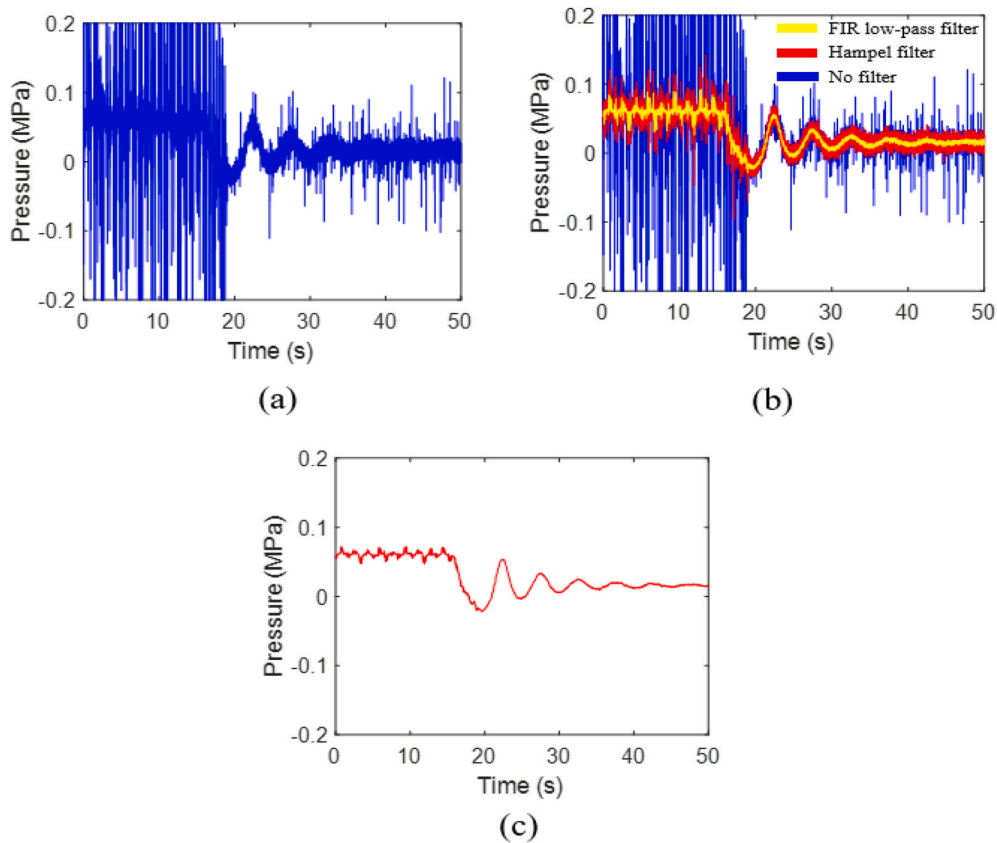


Fig. 6. Signal acquired in the laboratory and the signals after filtering: (a) signal acquired in the laboratory; (b) comparison of signal in the filtering process; (c) signal after comprehensive filtering.

amplitude attenuation were added at 13.33 s and 13.36 s, respectively. In this way, various pressure disturbances were added at a specified time to characterize the dynamic response time at two fractures. Finally, after the superposition structure and a certain amplitude treatment, the simulated signal representing the water hammer pressure wave signals characteristics is obtained. As a whole, the simulated signal of the water hammer pressure wave signal is attenuated (Fig. 8a). The thick arrow position indicates that the pressure begins to drop after the pump is stopped at 10 s. The enlarged part indicated by the dashed arrow represents the pressure drop caused by the water hammer pressure wave signals passing through multiple fractures. The corresponding abscissa is

13.33 s and 13.36 s, respectively, which is the pressure response time of multi-fracture fluctuation. The spectrum of the simulated signal is shown in Fig. 8b. The frequency is within 1 Hz, the main frequency is 0.1286 Hz, and the spectrum has a long tail. The one-dimensional cepstrum of the signal is shown in Fig. 8c. The pressure response obtained by cepstrum analysis are 13.33 s and 13.36 s, respectively, which are consistent with the pressure response of multi-fracture fluctuation constructed by the simulated signal above. Therefore, the cepstrum analysis can reasonably identify the pressure responses of multi-fracture.

The method of constructing the simulated signal with noise is as

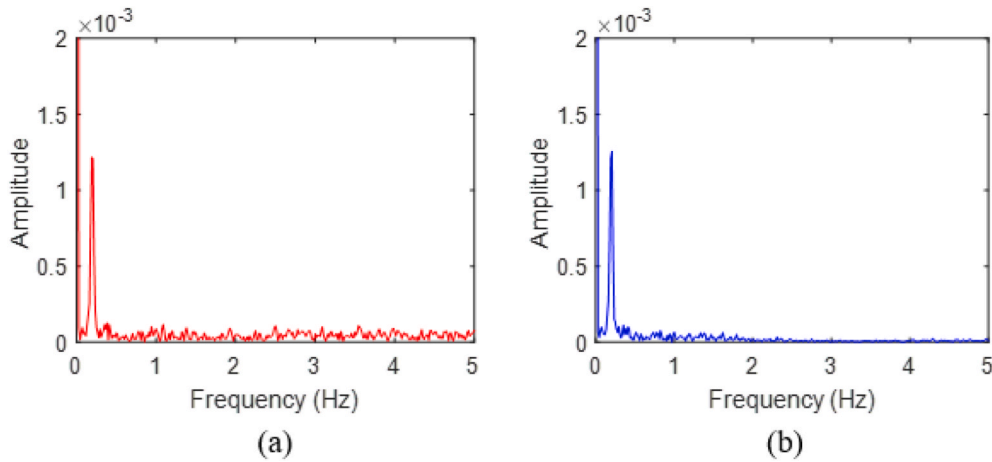


Fig. 7. Spectrum of the signal before and after filtering: (a) spectrum of the signal collected in the laboratory; (b) spectrum of the signal after comprehensive filtering.

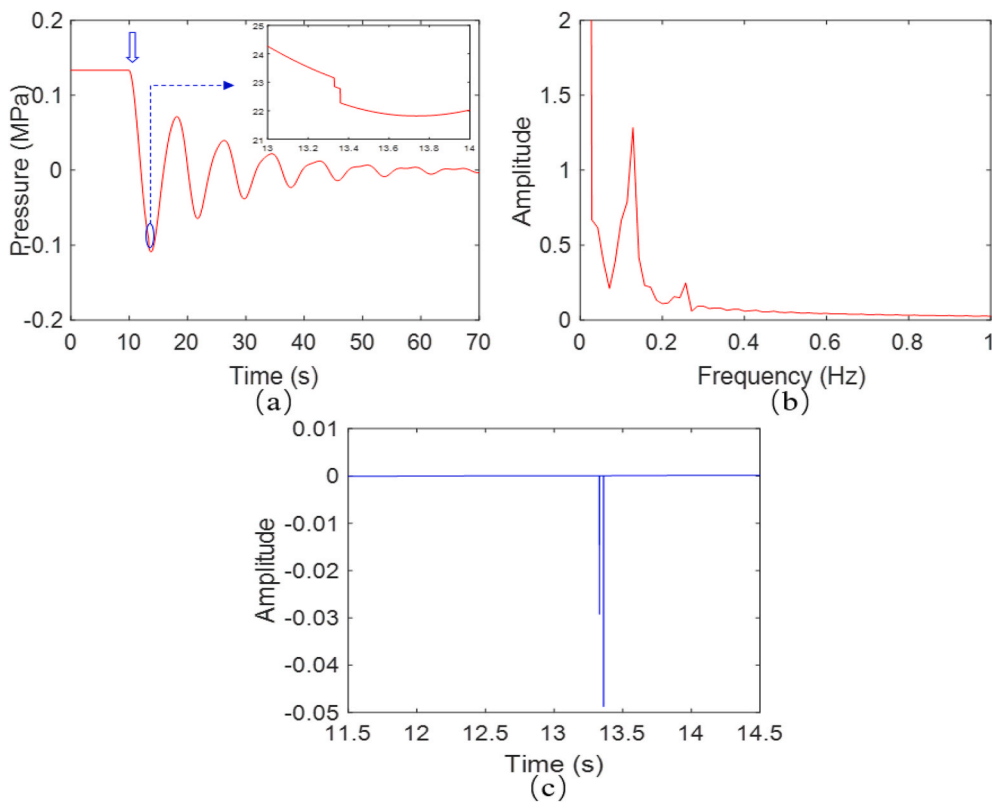


Fig. 8. Simulated signal with fractures, the signal spectrum and the signal cepstrum: (a) simulated signal with fractures; (b) spectrum of simulated signal with fractures; (c) cepstrum of simulated signal with fractures.

follows: the spike noises, random noises, low-frequency noises of fixed frequency which is close to the useful signal frequency, and high-frequency noises of fixed frequency which is much different from the useful frequency are added into the pure simulated signal with two fractures. The simulated signal containing noise is shown in Fig. 9a. We adopted the new comprehensive filtering model to filter the simulated signal. The red part in Fig. 9b is the first step to remove the spike noise by Hampel function. You can see that a lot of the spike noises have been filtered out without over-smoothing the overall curve. Fig. 9c shows the signal after adopting FIR low-pass filtering in the second step, and the high-frequency noises and random noises are partially suppressed. Fig. 9d is the signal after adaptive noise cancellation, that is, the signal after the comprehensive filtering. The noises are further filtered out, and

it can be seen that the pressure response of fracture is also preserved. And the pressure response of two fractures can be identified as 13.33 s and 13.36 s, respectively.

The comparison of SNR and MSE of signal with fractures before and after comprehensive filtering is shown in Fig. 10. Fig. 10a shows the SNR before and after the comprehensive filtering. The SNR before filtering is about 27.41844 dB. After Hampel filtering, the SNR is about 29.78164 dB. After FIR low-pass filtering, the SNR is about 32.63201 dB. After the comprehensive filtering, the SNR is about 34.21982 dB, and the SNR gain is about 6.80138 dB. Fig. 10b shows the MSE before and after the comprehensive filtering. The MSE before filtering is about 0.1047. After Hampel filtering, the MSE is about 0.0797. After FIR low-pass filtering, the MSE is about 0.0574. After the comprehensive filtering, the MSE is

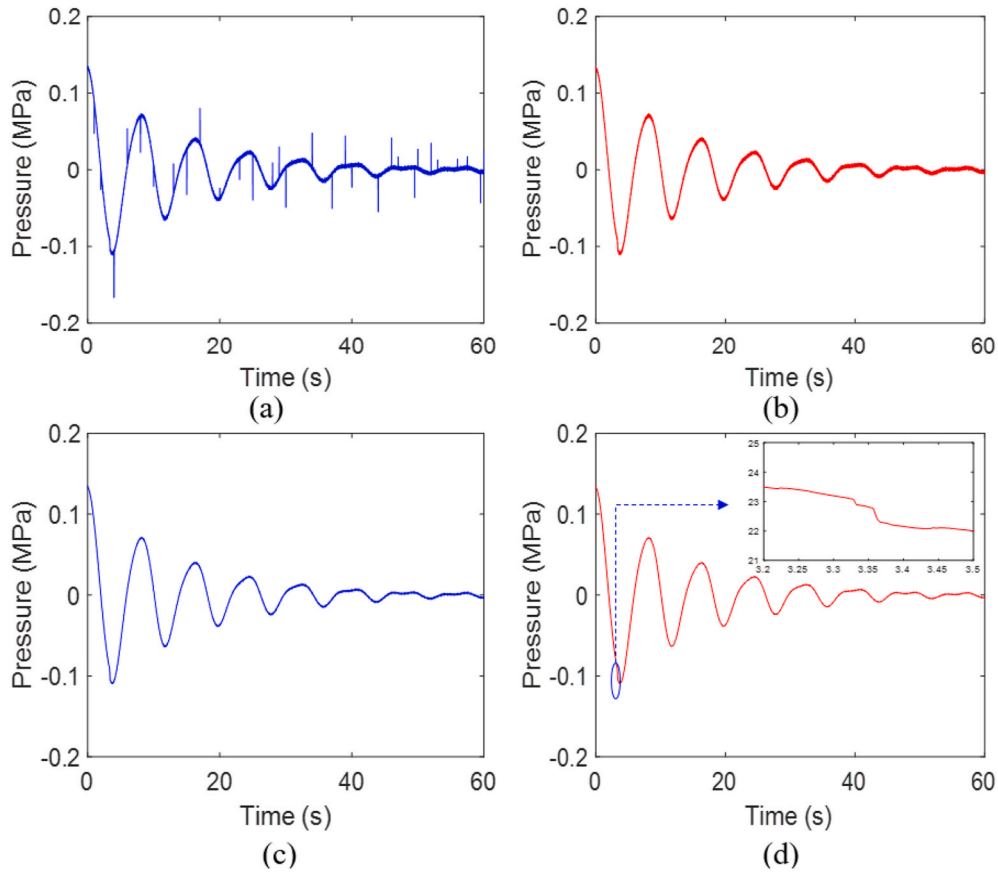


Fig. 9. Simulated signal with noise and the signals after filtering: (a) simulated signal with noise; (b) the signal after Hampel filtering; (c) the signal after FIR low-pass filtering; (d) the signal after adaptive noise cancellation.

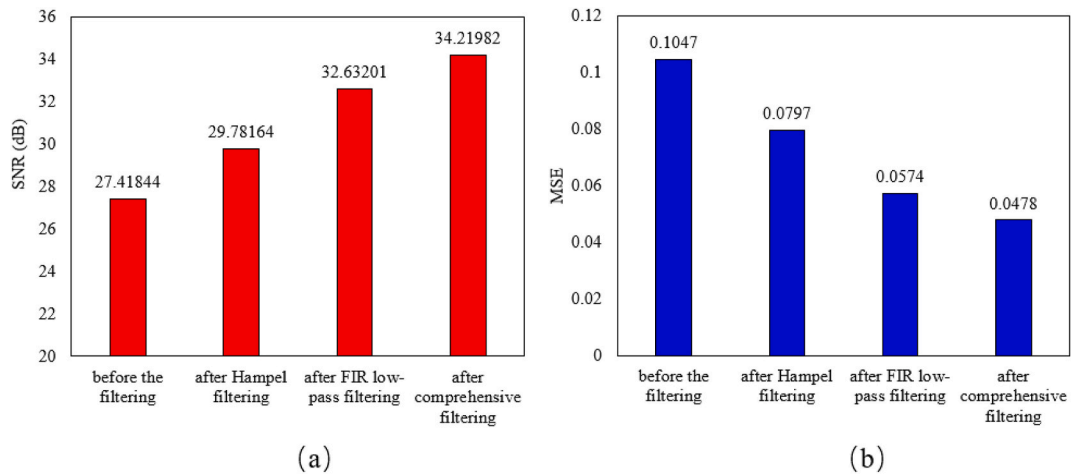


Fig. 10. Comparison of SNR and MSE before and after comprehensive filtering: (a) comparison of SNR before and after filtering; (b) comparison of MSE before and after filtering.

about 0.0478, and the MSE decreases about 0.0569. After the comprehensive filtering, the SNR of the water hammer pressure wave signal with fractures is improved, and the MSE is reduced. Moreover, the filtering effect is better than that of single Hampel filtering and FIR low-pass filtering.

The spectrum of the signal with fractures after comprehensive filtering is shown in Fig. 11a. The high frequency component is filtered out. And the main frequency of the water hammer pressure wave signal is retained, which is 0.1286 Hz. The one-dimensional cepstrum of the

signal is shown in Fig. 11b. The response times obtained by cepstrum analysis are 13.33 s and 13.36 s, which are consistent with the two pressure-response-times above.

4. Evaluation of noises effect

This model has a good effect on the water hammer pressure wave signals without fractures. Identifying response time should be taken into account when the comprehensive filtering model is used to filter the

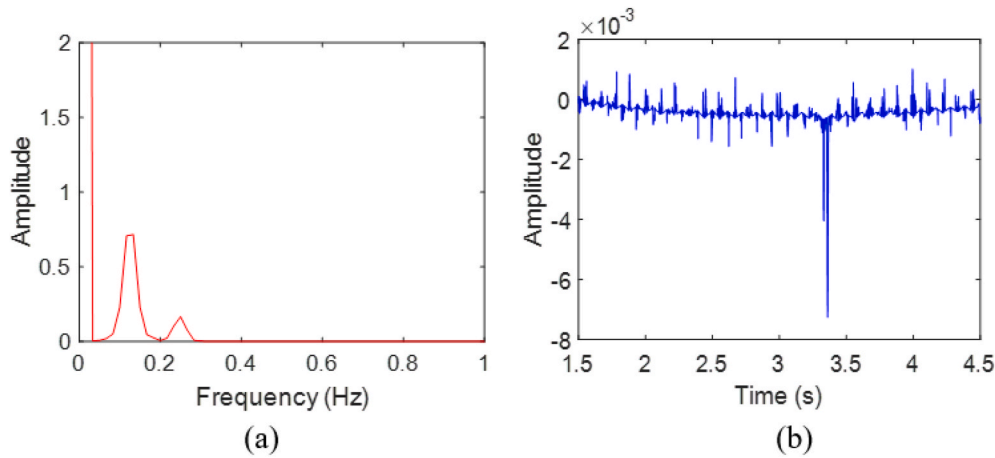


Fig. 11. Spectrum and cepstrum of the filtered signal with fractures: (a) spectrum of filtered signal with fractures; (b) cepstrum of filtered signal with fractures.

signal with fractures. The added noises will have different degrees of influence on the filtering effect, so it has certain limitations. If the added noises are too large, the pressure response of fracture will be completely submerged, then the pressure response after filtering will be challenging to identify. The influence of noise intensity on the model is discussed below, and the boundary of adding noise intensity is explained.

Effect of low-frequency noises. Because the useful frequency of the water hammer pressure wave signals is low frequency, low-frequency noises affect the overall shape of the signal but have little influence on the information at the pressure response. Adding too much low-frequency noises cause great distortion to the whole signal oscillation and reduce the SNR. When excessive low-frequency noises are added to

the water hammer pressure wave signals with fractures (Fig. 12a), the overall shape of the signal produces excellent distortion, but the pressure response time is not submerged at this time. After filtering, the pressure response of fracture (Fig. 12b) is basically preserved, but the overall distortion of the signal still exists. After analyzing the filtered signal cepstrum (Fig. 12c), the response time is 13.33 s and 13.36 s, respectively, which is consistent with the two pressure response times. Under this verification condition, the more extensive low-frequency noises only affect the signal overall shape and does not have much interference to the recognition of pressure response times.

Effect of high-frequency noises. High-frequency noises have a great influence on the pressure response times. The excessive high-

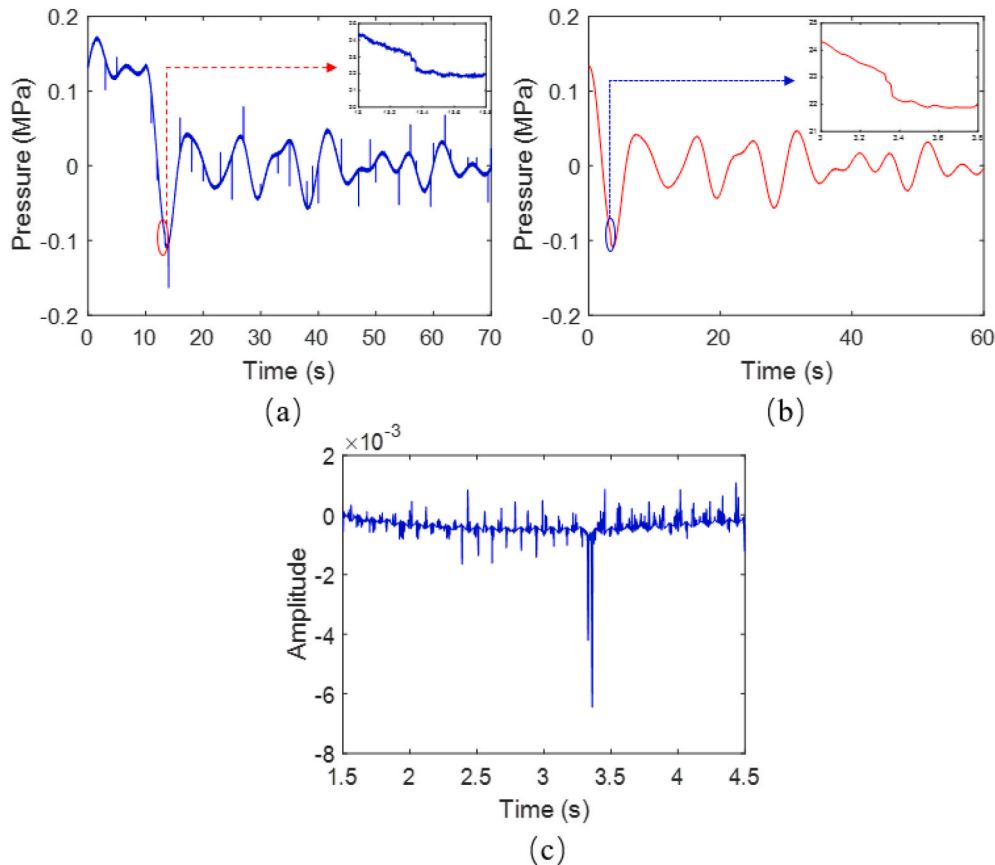


Fig. 12. The case after adding excessive low-frequency noises: (a) signal after adding excessive low-frequency noises; (b) signal after filtering out excessive low-frequency noises; (c) cepstrum of the signal after filtering out excessive low-frequency noises.

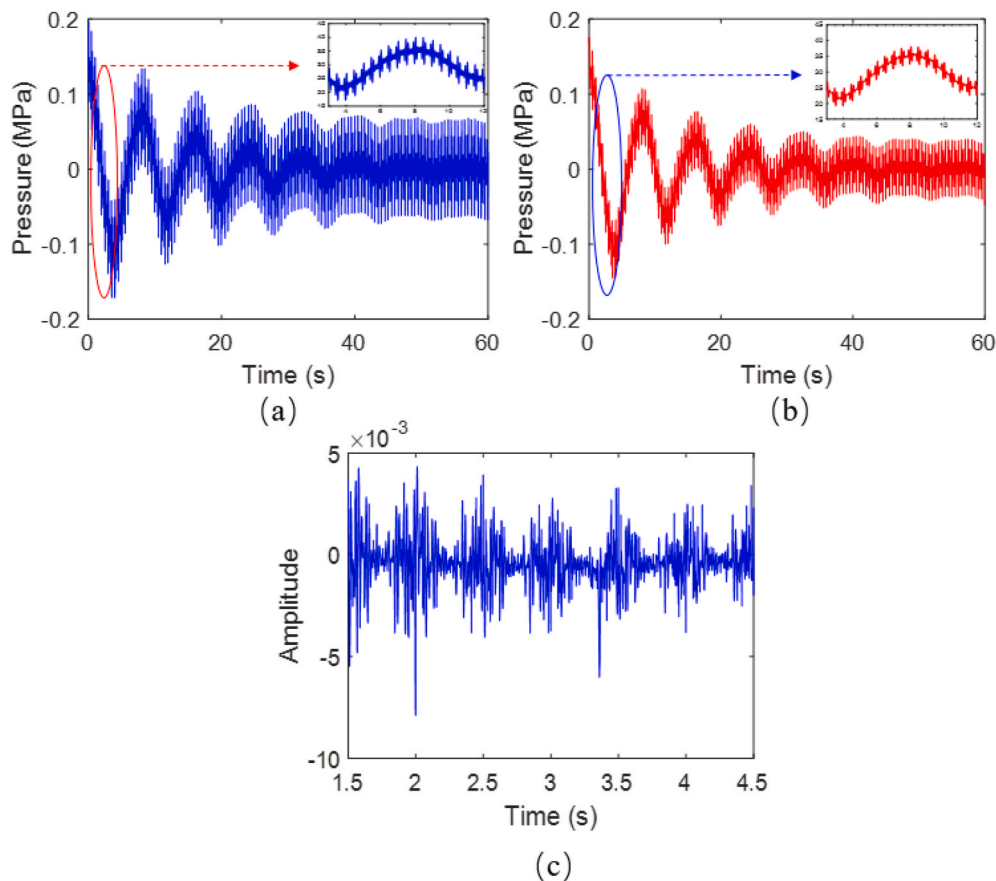


Fig. 13. The case after adding excessive high-frequency noises: (a) signal after adding excessive high-frequency noises; (b) signal after filtering out excessive high-frequency noises; (c) cepstrum of the signal after filtering out excessive high-frequency noises.

frequency noises will drown the information at the pressure response times. After filtering, useful information will be lost, and the pressure response location cannot be accurately identified. When excessive high-frequency noises are added to the water hammer pressure wave signal with fractures (Fig. 13a), the high-frequency noises in a certain period will produce more intensive oscillations, the pressure response of the fractures is completely submerged. After filtering, the pressure response of the fractures (Fig. 13b) has greatly interfered. It can't be recognized. By analyzing the signal cepstrum (Fig. 13c), the specific position of the pressure response time cannot be read out. Under this verification condition, the high-frequency noises produce too much interference to identify response time and even wholly drown the pressure response information.

Effect of random noises. Random noises will also have a great impact on the identification of pressure response time. Too much random noises can also drown out pressure response time. After filtering, the pressure response time will be lost, and the pressure response location cannot be accurately identified, thus affecting the judgment of the fracture location. When excessive random noises are added to the water hammer pressure wave signal with fractures (Fig. 14a), the pressure response time is entirely submerged by the random noises. After filtering, we cannot identify the pressure response of the fractures (Fig. 14b), and the pressure response time cannot be accurately read from the cepstrum of the signal (Fig. 14c). Under this verification condition, excessive random noises will also cause significant interference in identifying pressure response time or even completely drown the pressure response information.

Selection of noises amplitude. Under the condition of this model, we can obtain the limit of noise intensity through multiple tests. In the test, we set the amplitude of low-frequency noises and high-frequency

noises to change from 0.01 and increase 0.01 each time. Similarly, the standard deviation of random noises to change from 0.01 and increase 0.01 each time. After changing the range each time, we observed the effect before and after filtering until the signal failed to recognize the pressure response of the fractures after filtering. The limit of noise intensity is as follows: First, the maximum amplitude of low-frequency noises should be about 0.15. Otherwise, although the pressure response time will not be submerged and can be identified after filtering, the overall distortion of the signal will be too large, resulting in low SNR of the filtered signal. Second, the maximum amplitude of high-frequency noises should be about 0.2. Otherwise, the pressure response of the fractures will be drowned, resulting in unidentifiable information of the pressure response time after filtering. Third, random noises variance should be less than 0.15^2 . Otherwise, the pressure response time will be submerged, resulting in the pressure response of the fractures cannot be identified after filtering. Finally, since both the random noises and the high-frequency noises have a significant influence on the pressure response time, the random noises and the high-frequency noises cannot be greater than the two limits at the same time. Excessive random noises and high-frequency noises will drown the pressure response time, resulting in the loss of useful information after filtering.

5. Conclusions and limitation

A new comprehensive filtering model of Fixed Parameter Filter - Adaptive Filter is studied in this paper to eliminate the noises in the water hammer pressure wave signals, which is contains three parts: (1) Main input channel: A two-step filtering method for the main input signal; (2) Reference input channel: A MFAN-MF method for the noise reference input signal. (3) Adaptive noise cancellation between the main

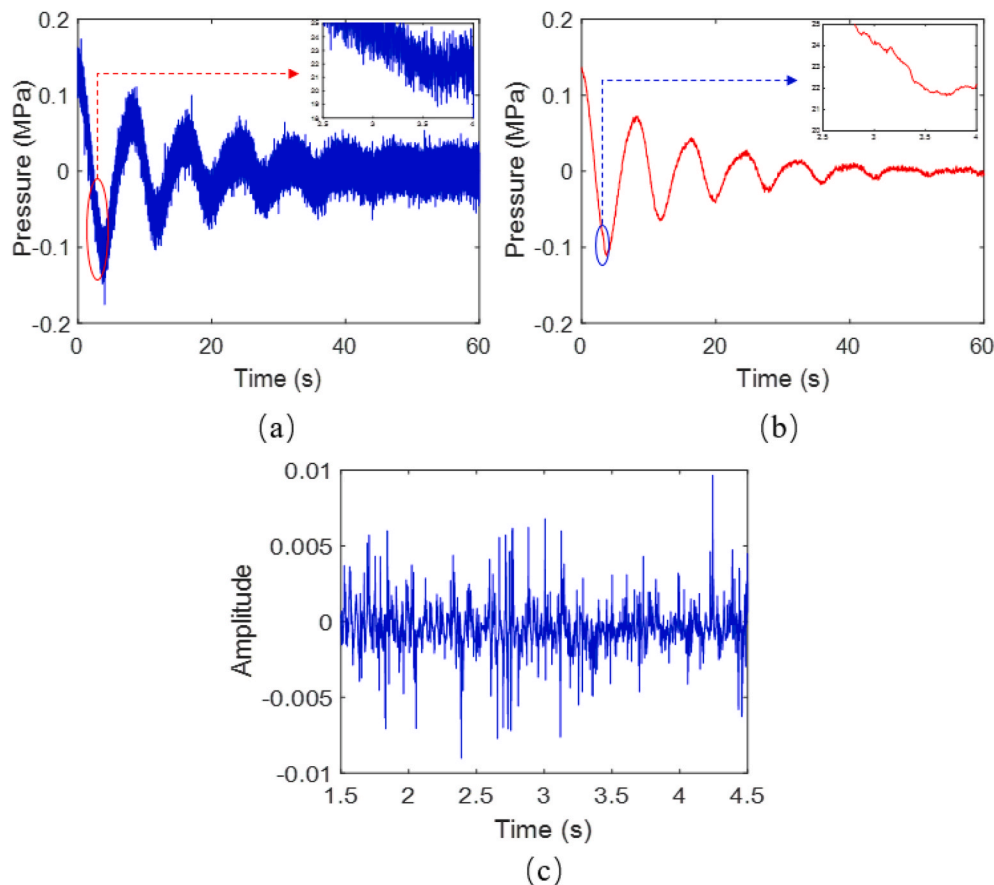


Fig. 14. The case after adding excessive random noises: (a) signal after adding excessive random noises; (b) signal after filtering out excessive random noises; (c) cepstrum of the signal after filtering out excessive random noises.

input signal and the reference input signal.

In the simulation test, this model is effective. For the water hammer pressure wave signals without fractures, the SNR gain is about 6.84429 dB, and the MSE can be reduced by about 0.0544. However, for the water hammer pressure wave signal with fractures, the SNR gain is about 6.80138 dB, the MSE can be reduced to about 0.0569, and the pressure response time can be well retained and identified. Signals collected in the laboratory are also used for verification. The results show that this model can retain the useful signals well and filter the noises.

This method has certain requirements for the noise intensity of the water hammer pressure wave signals with fractures. Within a certain range, the amplitude of low-frequency noise is about 0.15, the amplitude of high-frequency noise is about 0.2, and the variance of random noise is less than 0.15². Besides, the random noises and the high-frequency noises should not be greater than the two limits simultaneously.

Credit author statement

Yujiao Li: Conceptualization, Methodology, Data analysis, Visualization and Writing. **Xiaodong Hu:** Funding acquisition, Supervision, Conceptualization, Writing-Review and Editing. **Fujian Zhou:** Supervision. **Yang Qiu:** Validation. **Zhuolong Li:** Software. **Yinghao Luo:** Data Curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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